Online Resource 1 (Revised)

Supplementary material for: "Prefecture-Level Older-Adult Solo Household Rate and Large-Volume Infusion Therapy Utilization in Japan: An Ecological Study"

Machine-readable source data underlying every table and figure below, together with all analysis code, are provided in the openly available data and code repository cited in the main text's Data Availability statement.

Table S1. WBGT monitoring-site coverage per prefecture

Statistic	Value
N (prefectures)	47
Mean sites per prefecture	17.9
SD	22.7
Median	14.0
Minimum	5 (Kanagawa)
Maximum	163 (Hokkaido)

Official Ministry of the Environment WBGT observations were aggregated from all available monitoring sites per prefecture. Hokkaido's outlying site count reflects its large land area and is discussed as a limitation in the main text (prefecture-level climate averaging).

Table S2. G004 public-data availability audit summary

Table type (10th NDB Open Data)	Stratification available	Cross-tabulated by prefecture × age?
Sex/age marginal table	National, by sex and age group	No (no prefecture breakdown)
Prefecture marginal table	By prefecture (all ages combined)	No (no age breakdown)
Treatment-month marginal table	By calendar month	No (no prefecture or age breakdown)
Secondary-medical-area marginal table	By secondary medical area	No (no age breakdown)

Prefecture-by-age cross-tabulated G004 counts are not publicly available in the 10th NDB Open Data; only the separate marginal tables listed above exist. The complete file inventory and audit trail supporting this determination are provided in the accompanying repository.

Table S3. Sensitivity analysis using the 2020 Census population denominator

Model	Cumulative adjustment	Unstd. beta (HC3 95% CI)	p

O-A	Unadjusted	723.4 (377.5, 1069.2)	<0.001
O-B	+ ageing rate	310.2 (-87.0, 707.4)	0.126
O-C	+ ageing rate + hospital beds	76.5 (-340.2, 493.2)	0.719
O-D	+ ageing rate + hospital beds + WBGT≥28d	-31.8 (-536.8, 473.3)	0.902

The O-A point estimate (723.4) is closely comparable to the primary (2023-denominator) Model O-A estimate; inferential statistics differ because this analysis uses HC3 robust standard errors. Descriptive attenuation O-A→O-B: 57.1%.

Table S4. Alternative-exposure (denominator) comparison

Stage	Estimand	Unstd. beta (95% CI)	p	Std. beta (95% CI)	Partial R^2
Unadjusted	O — % of all households with older adult living alone	656.2 (326.9, 985.6)	<0.001	0.521 (0.260, 0.783)	0.272
Unadjusted	N — % of 65+ population living alone	0.3 (-249.8, 250.3)	0.998	0.0003 (-0.327, 0.328)	~0
Fully adjusted	O (O-D)	-9.4 (-493.9, 475.1)	0.970	-0.007 (-0.392, 0.377)	~0
Fully adjusted	N (analogous full adjustment)	-25.7 (-227.9, 176.4)	0.803	-0.034 (-0.298, 0.231)	0.001

Table S5. Sensitivity analyses (original exposure, O-D basis)

Analysis	N	Exposure coefficient	95% CI	p
O-D (reference)	47	-9.4	(-493.9, 475.1)	0.970
Beds → clinics (O-C)	47	55.6	(-304.9, 416.2)	0.762
Beds → clinics (O-D)	47	-35.6	(-361.7, 290.6)	0.831
Exclude Tokyo, Osaka, Kanagawa	44	-42.0	(-645.6, 561.6)	0.891
WBGT ≥31d substituted	47	96.2	(-304.6, 497.1)	0.638
WBGT ≥33d	47	-25.5	(-436.5, 385.5)	0.903

substituted				
Cumulative WBGT excess substituted	47	90.6	(-320.5, 501.7)	0.666

Leave-one-prefecture-out (47 iterations): exposure coefficient range -107.9 to 92.8; 95% CI crossed zero in all 47 iterations.

Influence diagnostics (O-D)

Cook's distance threshold (4/N) = 0.085. Hokkaido: Cook's D = 0.667 (leverage 0.431); Kochi: 0.190; Hiroshima: 0.144.

Nonlinearity (O-D)

Quadratic term: partial F p = 0.142, AICc 845.6 (linear) vs 845.9 (quadratic). Natural cubic spline (df=3): p = 0.141, AICc 845.9. See Figure S2 for the predicted curve with 95% confidence band.

Audit-only median-split reproduction (NOT a primary result)

High stratum (≥ median): β = -192.9 (95% CI -1060.4, 674.6), p = 0.647. Low stratum (< median): β = 20.7 (95% CI -973.9, 1015.3), p = 0.966. Neither stratum shows a significant association under full adjustment; this analysis does not support dose-response and reproduces the original manuscript's stratified approach only for audit purposes.

Table S6. Exploratory WBGT metrics with FDR correction

Heat metric	Nominal p (metric's own association with outcome)	BH-FDR-adjusted p
WBGT ≥31 days	0.977	0.977
WBGT ≥33 days	0.018	0.055
Cumulative WBGT excess (>28)	0.700	0.977

WBGT ≥33 days showed a nominal association that did not remain significant after FDR correction.

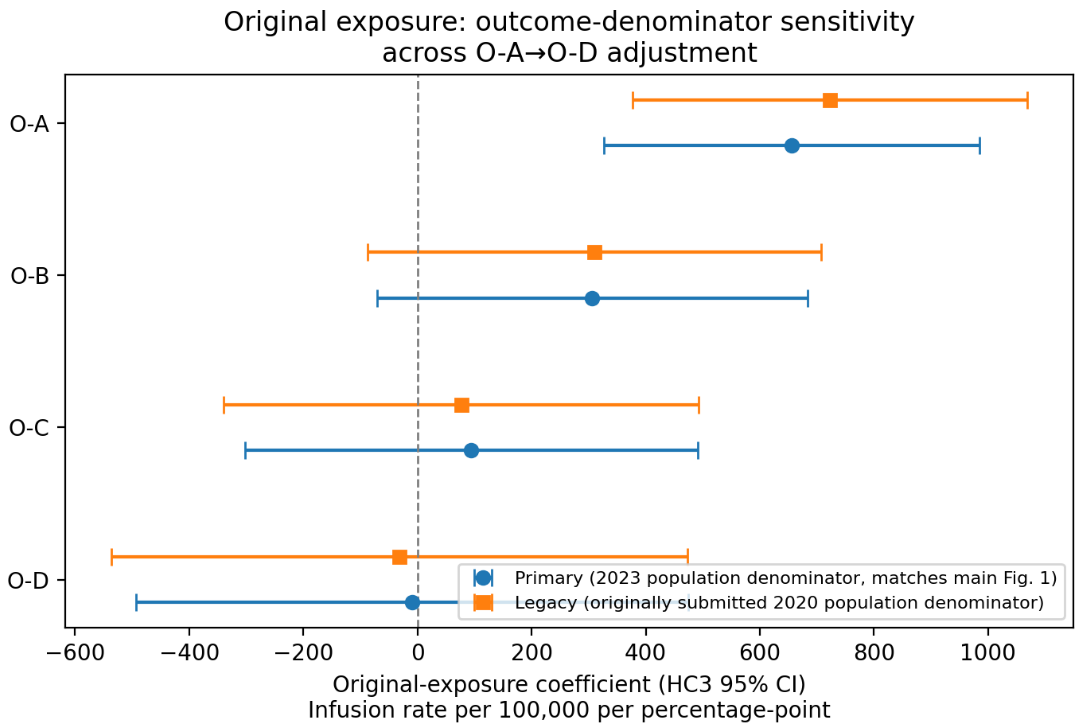
Table S7. Exploratory comparison-outcome analysis (formerly "negative control")

Outcome	Pearson r	R^2	Standardized β (95% CI)	p
Primary outcome (large-volume infusion therapy)	0.521	0.272	0.521 (0.260, 0.783)	<0.001
Comparison outcome	0.424	0.180	0.424 (0.088, 0.760)	0.013

(general outpatient utilization)				

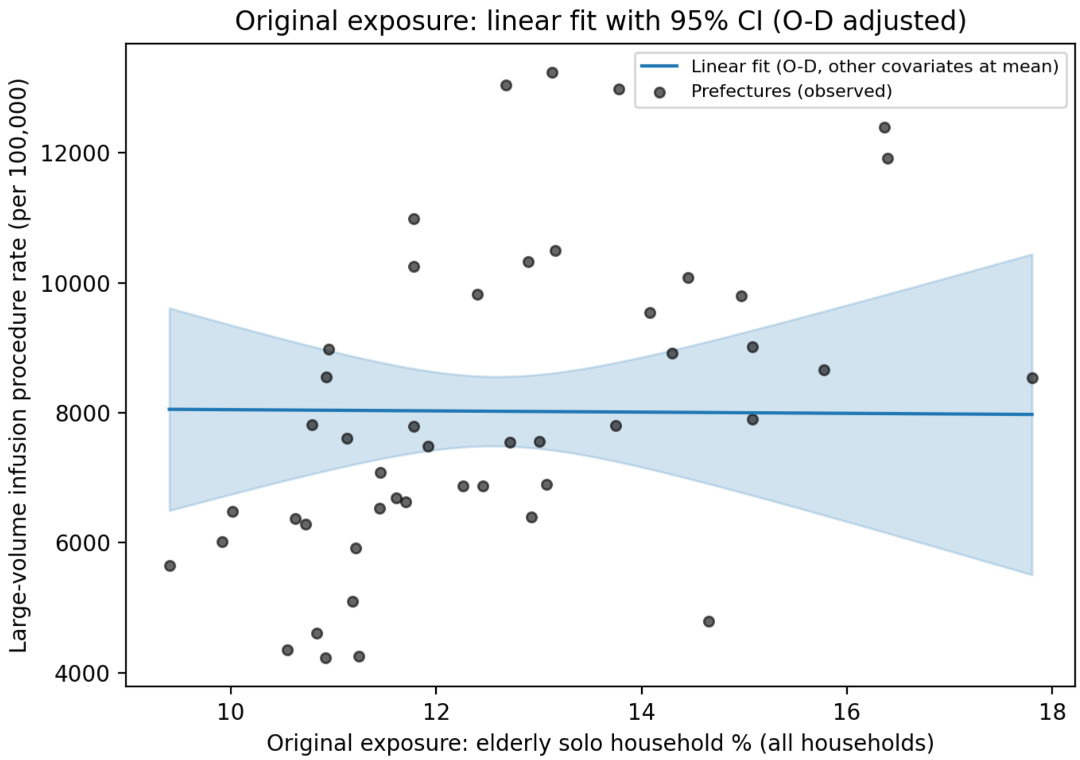
Reported under the primary (original) exposure, unadjusted (Model O-A basis); standardized comparisons only, consistent with Reviewer 2's comment 5. The standardized association with infusion therapy (0.521) is only modestly larger than with general outpatient utilization (0.424); this does not support the outcome specificity implied by the originally reported raw-coefficient ("sixfold") comparison, which has been removed.

Figure S1



Coefficient plot for the primary exposure across O-A–O-D with HC3 95% CIs (same as main Figure 1, reproduced here alongside the legacy-denominator estimates for comparison). Blue circles: primary analysis (2023 population denominator, matches main Figure 1). Orange squares: sensitivity analysis using the 2020 Census population denominator (Table S3).

Figure S2



Predicted-value curve (linear fit) with 95% confidence band across the observed range of the primary exposure, from the fully adjusted (O-D) model with the ageing rate, hospital beds, and WBGT≥28d covariates held at their mean values. Grey points show observed prefecture-level data (unadjusted).